

Sagedays Vienna: Integer Polytope Exercises

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Programming Exercises

Exercise. Fetch the polytopes from <https://github.com/gabrieleballetti/small-lattice-polytopes>. What is the maximum number of vertices that a lattice 3-polytope with (Euclidean) volume 1 can have?

Exercise. Define a function that receives a lattice polytope $P \subset \mathbb{R}^n$ as an input and returns the function

$$t_P: \mathbb{R}^n \rightarrow \mathbb{N}, \quad v \mapsto |(P - v) \cap \mathbb{Z}^n| \quad (1)$$

as an output.

Exercise The *support function* of a polytope $P \subset \mathbb{R}^n$ is defined as

$$h_P: \mathbb{R}^n \rightarrow \mathbb{R}, \quad h_P(u) = \max_{v \in P} \langle v, u \rangle.$$

Write a method that receives a polytope $P \subset \mathbb{R}^n$ (as an object of the class `Polyhedron`) as an input and returns the support function h_P as an output. Next, suppose that P is given by inequalities. Use the class `MixedIntegerLinearProgram` to define the support function *without* enumerating the vertices.

Exercise. Write a method that visualizes t_P (cf. (1)) for a lattice polygon $P \subset \mathbb{R}^2$. Maybe the package `HyperplaneArrangements` is helpful.

Theoretical Exercises

Exercise. Consider the polytope $P_0 = \text{conv}([0, 1]^n \cup [-1, 0]^n)$. Show that

$$P_0 = \{x \in \mathbb{R}^n : |x_i| \leq 1, |x_i - x_j| \leq 1, \forall 1 \leq i \leq j \leq n\}.$$

Exercise. For a permutation $\pi \in S_n$ consider the simplex

$$\Delta_\pi = \{x \in \mathbb{R}^n : 0 \leq x_{\pi_1} \leq \dots \leq x_{\pi_n} \leq 1\}.$$

Show that the simplices Δ_π , when translated by integer vectors, tessellate $\mathbb{R}^n$, i.e.,

$$\mathbb{R}^n = \bigcup_{z \in \mathbb{Z}^n} \bigcup_{\pi \in S_n} (\Delta_\pi + z),$$

and the union is disjoint up to sets of measure zero. Use this to prove $\text{vol} P_0 = n + 1$, where P_0 is as in the previous exercise.

Exercise. A *parity polytope* $P \subset \mathbb{R}^n$ is an origin symmetric lattice polytope that contains a lattice point of each parity class of $\mathbb{Z}^n/2\mathbb{Z}^n$. It is called a *strong* parity polytope, if its vertices are exactly its non-zero lattice points. Think of a parity polytope $P \subset \mathbb{R}^n$ which is not a strong parity polytope. You can consult Baletti's database for inspiration.

In the following, let t_P be defined as in (1).

Exercise. Verify that the map

$$T: \{\text{Lattice polytopes in } \mathbb{R}^n\} \rightarrow \{\mathbb{R}^n/\mathbb{Z}^n \rightarrow \mathbb{R}\}, \quad P \mapsto t_P$$

enjoys the following properties:

1. $T(P \cup Q) = T(P) + T(Q) - T(P \cap Q)$, whenever $P \cup Q$ and $P \cap Q$ are lattice polytopes.
2. $T(gP + z)(v) = T(P)(g^{-1}v)$, for any $g \in \text{GL}_n(\mathbb{Z})$ and $z \in \mathbb{Z}^n$.
3. $T(P) = \sum_{z \in \mathbb{Z}^n} \mathbf{1}[P - z]$, where $\mathbf{1}[X]$ denotes the 0-1-indicator function of a set $X \subset \mathbb{R}^n$.

Exercise. For a lattice polytope $P \subset \mathbb{R}^n$ show that

$$\text{vol}(P) = \int_{[0,1]^n} t_P(v) \, dv.$$

Exercise. Let $G_i(P)$ be the i th Ehrhart coefficient of a lattice polytope $P \subset \mathbb{R}^n$. Show that there exists a linear operator F_i defined on the function space $\{\mathbb{R}^n/\mathbb{Z}^n \rightarrow \mathbb{R}\}$ such that $G_i(P) = F_i(t_P)$.

Exercise. Let $P \subset \mathbb{R}^n$ be a lattice polytope and let $F \subset P$ be a facet of P . Can you compute t_F from t_P ?